

LZW Compression Question

Input Text:

A	G	A	G	A	G	A	C	T	C	T	G	T	C	A
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Tasks

1. Apply LZW compression to the input text shown above.
2. Show each dictionary expansion step until the dictionary reaches 13 entries.
3. Identify at least two repeating substrings in the input.
4. Provide the final LZW encoded output sequence.
5. Calculate:
 - Original size in bits (8 bits per symbol)
 - Compressed size in bits using the final dictionary size

Examiner Information (not shown to students):

Initial Dictionary: {'A': 0, 'C': 1, 'T': 2, 'G': 3}

Final Dictionary Size: 13

Original Bits: 120 bits

Compressed Bits: 40 bits

Encoded Output: [0, 3, 4, 6, 1, 2, 8, 3, 9, 0]

Step-by-Step LZW Compression Solution

w	c	wc	Output Code	New Dictionary Entry
	A	A		
A	G	AG	0	AG → 4
G	A	GA	3	GA → 5
A	G	AG		
AG	A	AGA	4	AGA → 6
A	G	AG		
AG	A	AGA		
AGA	C	AGAC	6	AGAC → 7
C	T	CT	1	CT → 8
T	C	TC	2	TC → 9
C	T	CT		
CT	G	CTG	8	CTG → 10
G	T	GT	3	GT → 11

T	C	TC		
TC	A	TCA	9	TCA → 12
A		A	0	

Final Encoded Output:

[0, 3, 4, 6, 1, 2, 8, 3, 9, 0]